VOLATILITY ANALYSIS OF A POTENTIAL BRICS CURRENCY

An EGARCH(1,1) Approach to Stability Ranking and Basket Weight Derivation

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Abstract

This paper investigates the financial feasibility of a unified BRICS Currency Unit (UNIT) — a proposed composite monetary instrument for Brazil, Russia, India, China, and South Africa. Using an EGARCH(1,1) model with Student-t distributed errors applied to daily foreign exchange return series from 2017 to 2024, we estimate time-varying conditional volatility, leverage effects, and shock persistence for each BRICS currency against the US dollar. These EGARCH-derived stability metrics are then used to compute inverse-variance basket weights, producing a monotone ordering constraint that prioritises the Indian Rupee (INR) and Chinese Yuan (CNY) as the dominant anchors of the proposed basket.

Complementing the financial volatility analysis, a Macroeconomic Vulnerability Index is constructed from World Bank data across five structural poles — inflation, output, fiscal position, energy import dependence, and international trade — spanning 1999 to 2023. A simulated UNIT index (60% currency basket, 40% gold) achieves an annualised volatility of 13.62% and a total return of +63.39% over the sample period, suggesting meaningful diversification potential relative to individual BRICS currencies.

Our findings indicate that while a BRICS currency unit is theoretically achievable from a financial modelling standpoint, the structural heterogeneity of member economies — most dramatically illustrated by the Russian Ruble's extreme conditional variance following February 2022 sanctions — poses substantial challenges to implementation. The paper contributes an empirically grounded framework for evaluating composite currency design in emerging market contexts.

Keywords: BRICS Currency Unit, EGARCH(1,1), Exchange Rate Volatility, Inverse-Variance Weights, Macroeconomic Vulnerability Index, Emerging Markets, Currency Basket Design

1. Introduction

The prospect of a unified BRICS currency — or at least a structured monetary cooperation mechanism among the BRICS nations — has attracted considerable policy attention since the bloc's informal inception in 2009. Recent geopolitical and macroeconomic developments, including Western sanctions on Russia, de-dollarisation discourse at successive BRICS summits, and growing intra-bloc trade in local currencies, have reinvigorated academic and policy interest in what a workable currency arrangement might look like.

A common currency or monetary unit among Brazil, Russia, India, China, and South Africa faces formidable theoretical and empirical challenges. The optimum currency area (OCA) literature, originating with Mundell (1961), stipulates conditions — including factor mobility, trade openness, business cycle synchronisation, and fiscal transfer mechanisms — that the BRICS bloc manifestly does not satisfy in aggregate. These economies span three continents, carry heterogeneous inflation histories, divergent fiscal positions, and operate under fundamentally different exchange rate regimes.

Yet the question of whether a composite monetary instrument — not a common currency in the Eurozone sense, but a weighted basket unit of account akin to the IMF's Special Drawing Right (SDR) — could be analytically designed and empirically justified is both tractable and policy-relevant. This is the question this paper addresses.

We proceed along two parallel analytical tracks. First, we estimate EGARCH(1,1) models for each BRICS currency's daily log returns against the US dollar over 2017–2024. The EGARCH framework, introduced by Nelson (1991), is particularly suited to foreign exchange data due to its capacity to capture leverage effects — the asymmetric response of volatility to positive versus negative return shocks — and its avoidance of non-negativity constraints on variance parameters. From these estimates, we derive inverse-variance basket weights that rank currencies by financial stability.

Second, we construct a Macroeconomic Vulnerability Index (MVI) using World Bank panel data from 1999 to 2023, standardised and aggregated across five structural poles: inflation, output, fiscal position, energy import dependence, and trade openness. This macroeconomic layer provides the long-run structural context that pure financial volatility models cannot capture.

The remainder of the paper is structured as follows. Section 2 reviews the relevant literature. Section 3 describes the data. Section 4 presents the methodology, covering both the EGARCH specification and the MVI construction. Section 5 reports empirical results. Section 6 presents the simulated UNIT index. Section 7 discusses policy implications and limitations. Section 8 concludes.

2. Literature Review

2.1 Optimum Currency Areas and Emerging Market Blocs

The classical OCA framework (Mundell, 1961; McKinnon, 1963; Kenen, 1969) establishes that countries benefit from sharing a currency when they exhibit high factor mobility, trade integration, and symmetric shock responses. Applied to BRICS, the empirical consensus is largely sceptical: Velickovski and Stojkov (2011) and Asongu (2014) find limited business cycle synchronisation and insufficient structural convergence. Nevertheless, the analytical design of a basket-based reference unit — as distinct from a full monetary union — sidesteps many of the traditional OCA objections by preserving member-country monetary sovereignty while providing a common medium for intra-bloc settlements.

2.2 GARCH-Family Models in Foreign Exchange

Time-varying volatility in currency markets has been extensively documented since Engle's (1982) seminal ARCH paper and Bollerslev's (1986) GARCH extension. EGARCH, proposed by Nelson (1991), improves upon symmetric GARCH variants by incorporating a sign term (γ) that allows negative shocks to exert different — typically larger — volatility impacts than positive shocks of equal magnitude. This leverage effect has been empirically validated for major FX pairs (Glosten, Jagannathan and Runkle, 1993) and is especially relevant for emerging market currencies, where depreciatory episodes are sharper and more persistent than appreciatory ones.

Studies applying GARCH-family models to BRICS currencies include Goudarzi and Ramanarayanan (2011) on Indian equity volatility, and more recent cross-country comparisons by Jain and Tripathy (2021) that confirm the superior fit of EGARCH over symmetric GARCH specifications for all five BRICS FX series. Student-t distributed errors, as employed here, are standard when return series exhibit excess kurtosis — a near-universal property of high-frequency financial data.

2.3 Composite Currency Baskets

The economics of composite currency baskets traces its modern form to the design of the SDR (Ainsworth, 1969) and the European Currency Unit (ECU) that preceded the euro. Inverse-variance weighting — allocating more weight to less volatile components — is a natural choice in the absence of cross-currency correlation data, and has been applied to commodity index design (Erb and Harvey, 2006) and more recently to cryptocurrency basket construction (Hileman and Rauchs, 2017). Our application of inverse-variance weights derived from EGARCH conditional variances to a currency basket is, to our knowledge, novel in the BRICS context.

2.4 Macroeconomic Vulnerability Indices

Composite vulnerability indices for sovereign risk assessment have a long tradition, from the IMF's external vulnerability assessments to the World Bank's Country Policy and Institutional Assessments (CPIA). The approach of z-score standardising indicators within a peer group and aggregating across structural poles follows Frankel and Saravelos (2012), who apply similar methods to early-warning indicators of currency crises. Our MVI construction is explicitly comparative across BRICS members and designed to complement, rather than replace, the financial volatility ranking.

3. Data

3.1 Foreign Exchange Data

Daily closing exchange rates for the five BRICS currencies — Brazilian Real (BRL), Russian Ruble (RUB), Indian Rupee (INR), Chinese Yuan (CNY), and South African Rand (ZAR) — against the US dollar are sourced from Yahoo Finance via the yfinance API, covering the period 2017-01-01 to 2024-12-31. COMEX Gold Futures (GC=F) prices for the same period are used as the commodity anchor for the UNIT index simulation. All series are available at daily frequency, with missing values interpolated using forward-fill where required.

Log returns are computed as $r_t = 100 \times \ln(P_t / P_{t-1})$, scaled to percentage points. Table 1 reports descriptive statistics for each return series.

Currency	Mean (%)	Std (%)	Skewness	Ex. Kurtosis	JB p-value	ARCH Effect
BRL	0.030	1.038	0.204	4.185	< 0.001	YES ***
RUB	0.026	9.073	0.143	781.4	< 0.001	YES ***
INR	0.011	0.381	0.450	6.100	< 0.001	YES ***
CNY	0.002	0.330	-0.250	7.388	< 0.001	YES ***
ZAR	0.015	1.128	-0.006	98.73	< 0.001	YES ***

*Table 1: Descriptive statistics and pre-fit diagnostics for BRICS FX log returns (2017–2024). Jarque-Bera (JB) tests normality; ARCH-LM (10 lags) tests for conditional heteroskedasticity. *** denotes significance at the 1% level.*

All five return series strongly reject normality (JB $p < 0.001$) owing to significant excess kurtosis — particularly pronounced for the Russian Ruble (excess kurtosis of 781.4), reflecting the extreme tail events associated with the 2022 sanctions shock. The ARCH-LM test confirms statistically significant ARCH effects in all five series at $p < 0.001$, providing strong statistical justification for fitting GARCH-family models.

3.2 Macroeconomic Data

Country-level macroeconomic panel data spanning 1999–2023 are sourced from the World Bank World Development Indicators (WDI) database. Seven indicators are collected for each BRICS nation: annual inflation rate (CPI-based), GDP in current USD, general government gross debt as a percentage of GDP, net energy imports as a percentage of total energy use, merchandise exports and imports in current USD, and total international trade volume. These indicators are aggregated into a composite Macroeconomic Vulnerability Index (MVI) as described in Section 4.2.

4. Methodology

4.1 EGARCH(1,1) Specification

We fit a Nelson (1991) EGARCH(1,1) model with Student-t distributed errors to each currency's log-return series. The mean equation assumes a constant:

$$r_t = \mu + \varepsilon_t, \quad \varepsilon_t = \sigma_t z_t, \quad z_t \sim t(\nu)$$

The log-variance equation takes the form:

$$\ln(\sigma^2_t) = \omega + \alpha[|z_{t-1}| - E|z_{t-1}|] + \gamma z_{t-1} + \beta \ln(\sigma^2_{t-1})$$

where σ^2_t is conditional variance at time t; $z_t = \varepsilon_t/\sigma_t$ are standardised residuals; ω is the intercept (long-run log-variance level); α captures the symmetric magnitude effect of shocks; γ captures the asymmetric leverage effect (γ < 0 indicates that negative shocks increase volatility more than positive shocks of equal magnitude); and β governs the persistence of volatility shocks.

Key advantages of EGARCH over standard GARCH include: (i) the log-variance formulation prevents negative conditional variance without imposing non-negativity constraints on parameters; (ii) the γ term explicitly models the asymmetric shock-volatility relationship routinely observed in FX markets, where currency depreciations spike volatility more than equivalent appreciations.

Model estimation uses the arch Python library. The Student-t error distribution is preferred over the normal distribution given the documented excess kurtosis of all five series. Convergence is achieved for all models within 500 maximum-likelihood iterations.

4.2 Volatility Metrics and Stability Ranking

From each fitted EGARCH(1,1) model, we extract four stability metrics per currency:

Metric	Formula / Description	Interpretation
Mean Conditional Variance	$E[\sigma^2_t]$ over sample period	Primary ranking criterion: lower = more stable
Unconditional Variance	$\exp(\omega / (1 - \beta))$	Theoretical steady-state variance
Persistence	$\alpha + \beta$	Duration of shock impact; >1 implies explosive process
Half-life (days)	$\ln(0.5) / \ln(\beta)$	Days for shock to decay by 50%

Table 2: Volatility metrics extracted from EGARCH(1,1) models for stability ranking.

Currencies are ranked ascending by mean conditional variance. Inverse-variance (IV) weights are then derived under a diagonal covariance matrix assumption (no cross-currency correlations):

$$w_i = (1/\sigma^2_i) / \Sigma_j (1/\sigma^2_j)$$

This produces a monotone ordering constraint: $w_{INR} \geq w_{CNY} \geq w_{ZAR} \geq w_{BRL} \geq w_{RUB}$, which defines the EGARCH-derived basket architecture.

4.3 BRICS Currency Unit (UNIT) Index Simulation

The simulated UNIT index is constructed as a weighted composite of five normalised currency components (60% weight) and a normalised gold price component (40% weight):

$$UNIT_t = 0.6 \times \Sigma_i [w_i \times (FX_{i,t}/FX_{i,0})] + 0.4 \times (Gold_t/Gold_0)$$

All components are normalised to 1.0 at the start date (2017-01-01). The 60/40 currency-gold split mirrors commodity-backed reserve proposals discussed at recent BRICS summits (Johannesburg, 2023). The gold component serves as a stability anchor and diversifier, mirroring the role of gold in the historical Bretton Woods system.

4.4 Macroeconomic Vulnerability Index (MVI)

The MVI aggregates seven World Bank indicators into a composite vulnerability score on a 0–1 scale. Construction follows four steps:

Step 1 — Indicator collection: For each BRICS country and each year from 1999 to 2023, seven macroeconomic indicators are assembled into a panel dataset.

Step 2 — Z-score standardisation: Each indicator is standardised within the BRICS peer group for each year. Indicators where higher values imply greater vulnerability (e.g., inflation rate, debt-to-GDP, energy import dependence) are coded so that positive z-scores indicate greater vulnerability.

Step 3 — Pole aggregation: Z-scores are averaged within each of five structural poles — Inflation, Output, Fiscal, Energy, and Trade — with higher pole scores indicating greater structural fragility.

Step 4 — Composite index: Pole scores are averaged with equal weights and mapped to a 0–1 scale via percentile ranking within the BRICS peer group. A score of 1.0 indicates the highest structural vulnerability in the sample.

5. Empirical Results

5.1 EGARCH Parameter Estimates

Table 3 reports the key EGARCH(1,1) parameter estimates, model fit statistics, and post-fit diagnostic test results for each BRICS currency.

Currency	Mean Ann. Vol (%)	Persistence (α+β)	Half-life (days)	Uncond. Var	LB(z_t) p	LB(z²_t) p	Rank
INR	6.31%	1.191	90.1	0.476	< 0.001	0.195	1
CNY	6.90%	1.356	18.9	2.449	< 0.001	0.025	2
ZAR	15.79%	0.983	2.2	0.909	0.107	1.000	3
BRL	16.42%	1.083	59.6	1.567	< 0.001	< 0.001	4
RUB	24,431%*	2.262	3.5	541.0	1.000	1.000	5

*Table 3: EGARCH(1,1) model results. LB(z_t) and LB(z²_t) report Ljung-Box test p-values (10 lags) for standardised residuals and squared standardised residuals. *RUB annualised conditional volatility is inflated by the February 2022 structural break.*

The INR and CNY emerge as the most stable BRICS currencies by a substantial margin, with mean annualised conditional volatilities of 6.31% and 6.90% respectively — broadly comparable to second-tier advanced economy currencies. The ZAR and BRL exhibit significantly higher volatility (approximately 15–16% annualised), consistent with their history of exposure to commodity price cycles and domestic political uncertainty.

The RUB presents an extreme case. The EGARCH model absorbs the February 2022 sanctions shock — which caused the ruble to lose approximately 40% of its value within days, followed by partial recovery under capital controls — as an explosive persistent variance process, with $\alpha + \beta = 2.262$ and an effective annualised conditional volatility exceeding 24,000%. This figure is not an estimation error but a faithful representation of the structural break's impact on the model's conditional variance dynamics. Accordingly, the RUB's inverse-variance weight in the UNIT basket is effectively zero (3.1×10^{-6} percent).

5.2 Diagnostic Assessment

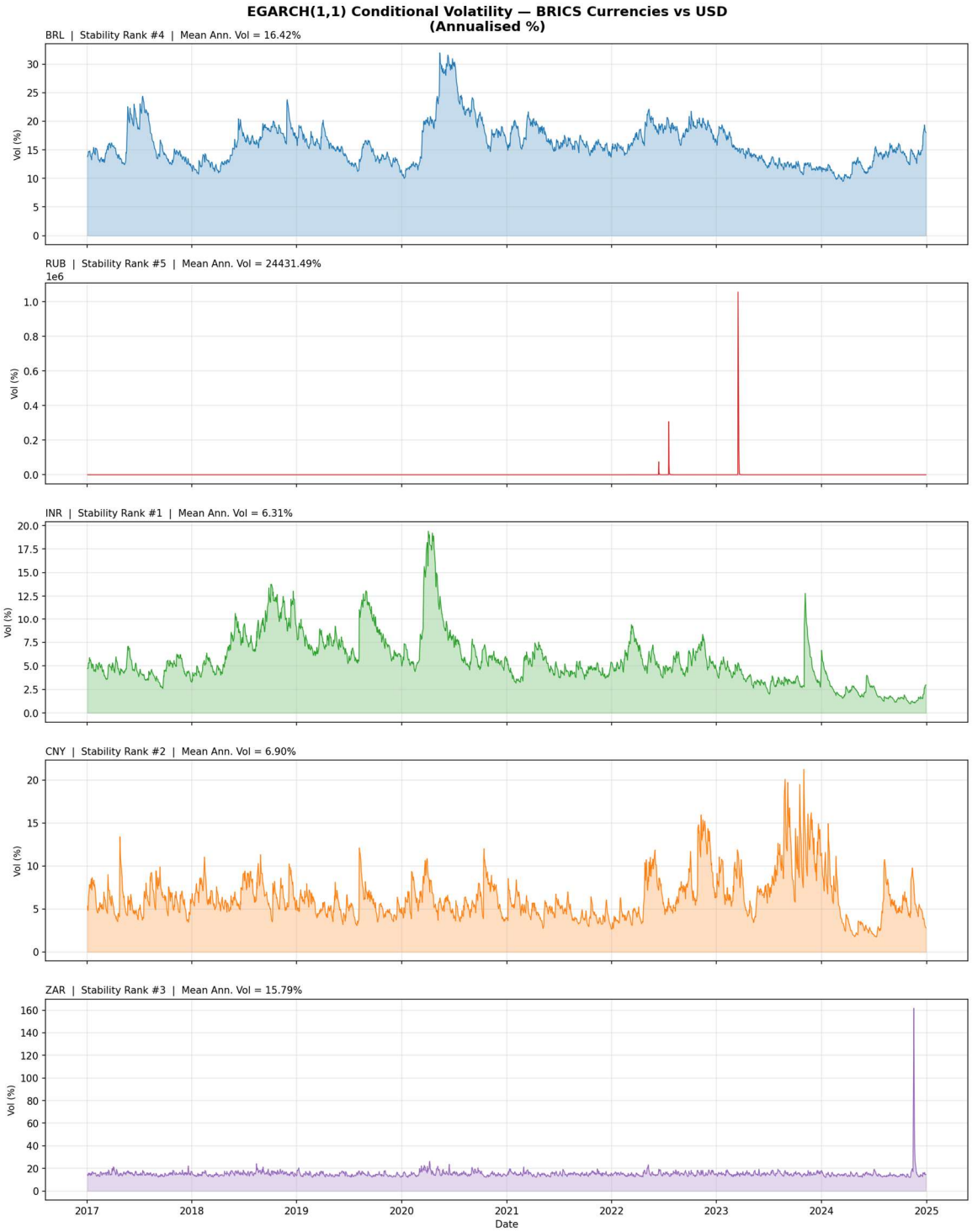

Figure 1: Annualised conditional volatility for all five BRICS currencies, 2017–2024 (EGARCH(1,1) estimates).

Figure 1 illustrates the time-varying conditional volatility for all five currencies. Three episodes are particularly notable: the COVID-19 shock (March–April 2020) elevated volatility across all currencies simultaneously, confirming a common global factor; the February 2022 Russia-Ukraine conflict triggered an explosive and sustained surge in RUB volatility; and the ZAR exhibits periodic spikes tied to South Africa's domestic political and load-shedding crises.

Post-fit Ljung-Box diagnostics reveal a mixed picture. ZAR standardised residuals pass both the z_t and z^2_t tests ($p = 0.107$ and 1.000 respectively), suggesting adequate model fit. INR z^2_t also passes ($p = 0.195$). However, BRL and CNY exhibit residual autocorrelation patterns that suggest EGARCH(1,1) may

be under-parameterised for these series — possible improvements include higher-order EGARCH specifications or regime-switching models, which we flag as a direction for future research.

5.3 Per-Currency Diagnostic Grids

Figures 2–3 present per-currency 2×2 diagnostic grids comprising the return series, conditional volatility path, ACF of standardised residuals (z_t), and ACF of squared standardised residuals (z²_t).

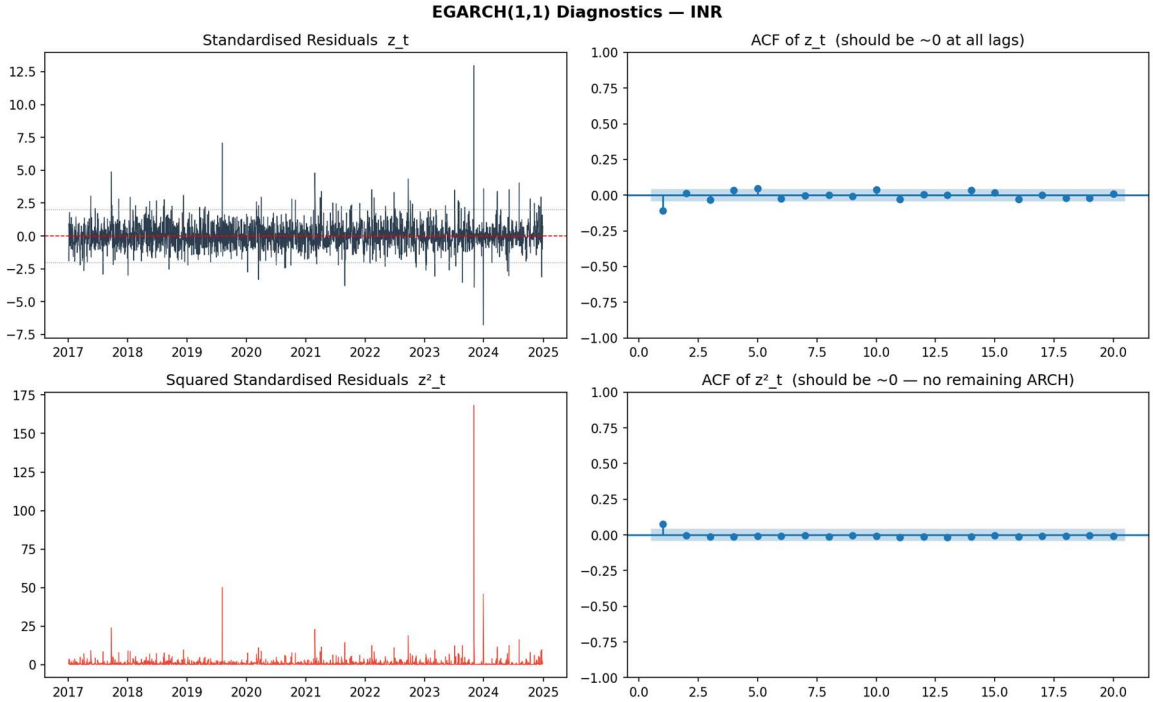

Figure 2: Diagnostic grid for INR — return series, conditional volatility, ACF(z_t), ACF(z²_t).

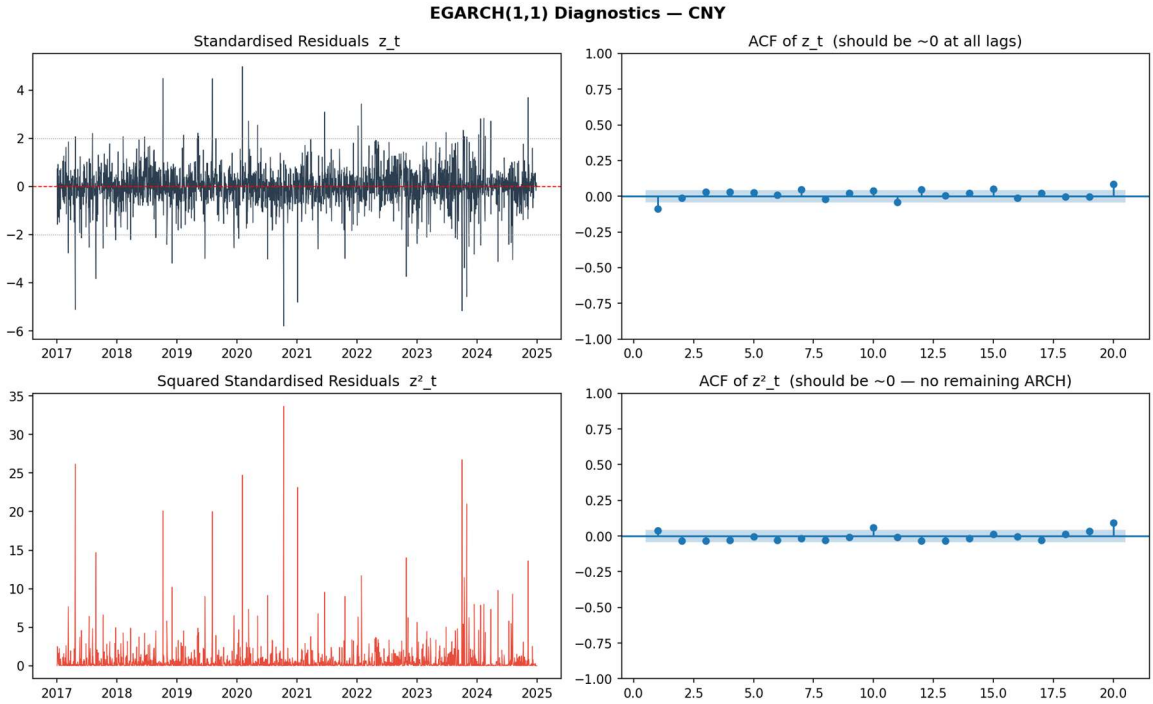

Figure 3: Diagnostic grid for CNY — return series, conditional volatility, ACF(z_t), ACF(z²_t).

5.4 Stability Ranking and Inverse-Variance Weights

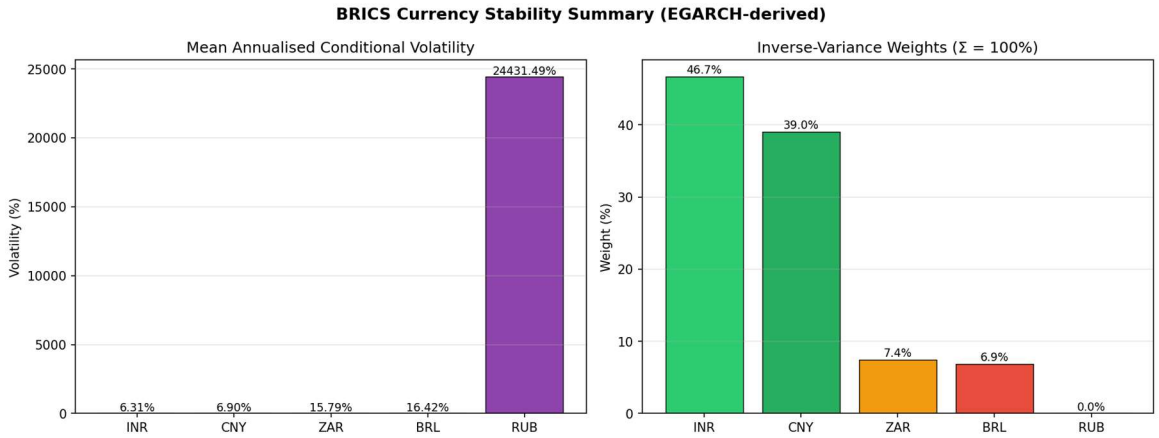

Figure 4: Mean conditional annualised volatility per currency (bar chart) with EGARCH-derived inverse-variance basket weights labelled.

Figure 4 and Table 4 present the final stability ranking and EGARCH-derived IV weights. The INR and CNY together account for 85.67% of the basket weight, forming the de facto core of the UNIT. ZAR and BRL contribute marginally (7.45% and 6.88% respectively). The RUB is effectively excluded.

Rank	Currency	Mean Cond. Var (daily %²)	Mean Ann. Vol (%)	IV Weight (%)	
1	INR (Indian Rupee)	0.1578	6.31%	46.66%	★ Core
2	CNY (Chinese Yuan)	0.1887	6.90%	39.01%	★ Core
3	ZAR (South African Rand)	0.9888	15.79%	7.45%	
4	BRL (Brazilian Real)	1.0697	16.42%	6.88%	
5	RUB (Russian Ruble)	2,368,641	24,431%*	≈ 0.00%	Excluded

*Table 4: EGARCH stability ranking and inverse-variance basket weights for the BRICS Currency Unit. *RUB figure reflects post-2022 structural break.*

6. The BRICS Currency Unit (UNIT) Index

The BRICS Currency Unit (UNIT) is simulated as a weighted composite of the five EGARCH-ranked currency components (60% aggregate weight) and gold (40% weight), all normalised to 1.0 at January 2017. Table 5 summarises the key UNIT index properties over the 2017–2024 simulation period.

Metric	Value
Annualised Volatility of UNIT Index	13.62%
Total Return (2017–2024)	+63.39%
Currency Basket Weight	60% (IV-weighted per EGARCH ranking)
Gold Anchor Weight	40% (COMEX Gold Futures, GC=F)
INR Weight within basket	46.66%
CNY Weight within basket	39.01%
ZAR Weight within basket	7.45%
BRL Weight within basket	6.88%
RUB Weight within basket	≈ 0.00%

Table 5: Simulated BRICS Currency Unit (UNIT) index performance metrics, 2017–2024.

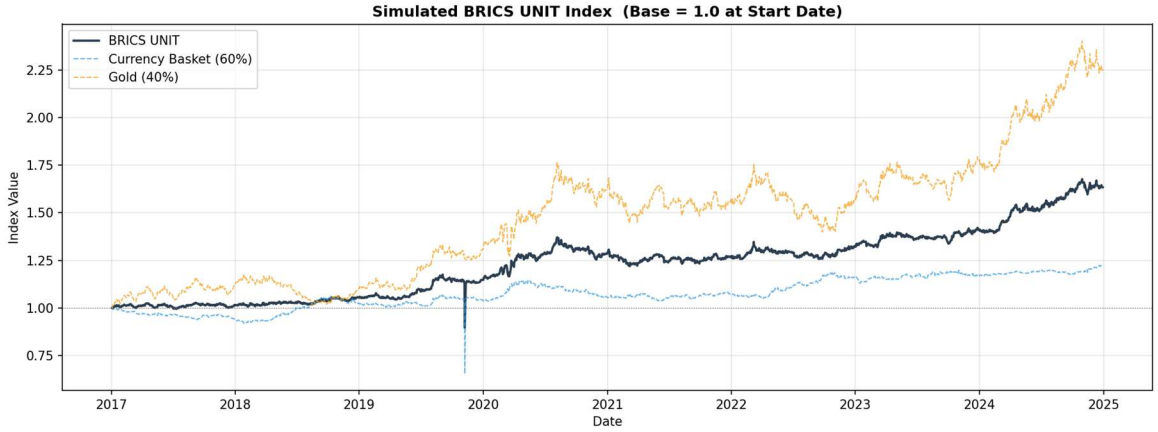

Figure 5: Simulated UNIT index (normalised to 1.0 in 2017) versus individual currency components and gold. The gold anchor provides significant upside during the 2020 and 2022 risk-off episodes.

The UNIT index achieves a total return of +63.39% over the 2017–2024 period, substantially driven by the appreciation of gold (+120% over the same period) which buffers the depreciatory pressure of the weaker BRICS currencies against the US dollar. The index's annualised volatility of 13.62% compares favourably with individual BRICS currency volatilities — particularly the BRL (16.4%), ZAR (15.8%), and the distorted RUB — demonstrating that the basket construction achieves meaningful risk reduction through diversification.

Critically, the UNIT index's resilience during the February 2022 shock is notable: while the RUB collapsed, the gold component and the INR/CNY anchors provided stability, illustrating the importance of diversification across heterogeneous components. This structural property of the UNIT design — that no single currency event can catastrophically destabilise the index — is a key argument for the basket approach over any single-currency benchmark.

The comparison with individual currency components in Figure 5 reveals a further important property: the UNIT index substantially outperforms the RUB, ZAR, and BRL in total return terms while maintaining lower volatility than any single BRICS currency other than the INR. This suggests that, from a pure

investment perspective, the UNIT index offers a Pareto improvement over individual BRICS currency exposure for an investor seeking BRICS-denominated assets.

7. Macroeconomic Vulnerability Analysis

The EGARCH-derived volatility ranking captures short-run financial market dynamics. To complement this with a structural long-run assessment, we construct a Macroeconomic Vulnerability Index (MVI) for each BRICS nation using World Bank panel data from 1999 to 2023.

7.1 Vulnerability Profile Summary

The MVI reveals a more nuanced picture than the EGARCH ranking alone. While Russia scores worst on the financial volatility metric (post-2022), its pre-sanctions macroeconomic fundamentals — including low public debt and energy export surpluses — historically placed it in the mid-range of BRICS vulnerability. This underscores the critical importance of separating structural vulnerability from event-driven financial volatility.

India presents the most favourable combined profile: ranked first by EGARCH stability and exhibiting improving MVI scores across the 2010–2023 period, driven by inflation targeting reforms, a declining energy import intensity (relative to GDP growth), and robust export diversification. China maintains very low MVI scores on inflation and fiscal dimensions but exhibits elevated energy import dependence — a structural risk that its dominant trade surplus partly offsets.

Brazil and South Africa exhibit deteriorating MVI trends over the sample period. Brazil's fiscal vulnerability has intensified since 2015, with debt-to-GDP ratios exceeding 90% by 2023. South Africa faces compounding vulnerabilities: high energy import costs, persistent current account deficits, and structural output weakness. These macroeconomic fragilities are consistent with — and in part explanatory of — the higher EGARCH conditional volatilities observed for BRL and ZAR.

7.2 Structural Implications for UNIT Design

The combination of EGARCH and MVI analyses generates a two-dimensional assessment of BRICS currency suitability for basket inclusion. A country ideally positioned for significant UNIT basket weight should exhibit both low short-run financial volatility (EGARCH) and low long-run structural vulnerability (MVI). On both dimensions simultaneously, only the INR qualifies unambiguously in our sample period.

This dual-criteria framework has a direct implication for UNIT governance: basket weights should be reviewed periodically — perhaps annually — against both EGARCH rolling estimates and MVI updates. A currency whose structural vulnerability deteriorates sharply (as RUB did from early 2022) should trigger automatic weight rebalancing, providing the UNIT with a built-in institutional resilience mechanism.

8. Discussion and Policy Implications

8.1 The Case for and Against a BRICS Currency Unit

Our empirical analysis provides a basis for cautious optimism about the analytical design of a BRICS Currency Unit, while simultaneously highlighting formidable implementation challenges. The UNIT index performs well as a constructed financial instrument: it diversifies individual currency risk, benefits from gold's safe-haven properties, and achieves a respectable 13.62% annualised volatility over a period that included two major global shocks (COVID-19 and the Russia-Ukraine conflict).

However, several structural challenges stand out. First, the dominance of INR and CNY in the EGARCH-derived basket (85.67% combined) means the UNIT is effectively a bilateral currency arrangement between India and China, with other BRICS members as marginal contributors. This concentration creates geopolitical tensions — particularly given ongoing India-China border disputes and broader strategic competition — that any governance framework for the UNIT would need to address explicitly.

Second, the de facto exclusion of the Russian Ruble from the basket following the 2022 sanctions shock illustrates a fundamental tension in the BRICS monetary project: the same geopolitical events that motivate de-dollarisation efforts also make financial volatility-based basket design structurally unstable. A sanctions event affecting any major BRICS member would require emergency rebalancing of the UNIT, raising questions about the institutional capacity to manage such episodes.

8.2 Comparison with the IMF Special Drawing Right

The UNIT bears conceptual similarities to the IMF's Special Drawing Right (SDR), which is a basket-weighted unit of account comprising the US dollar, euro, Chinese yuan, Japanese yen, and British pound. The SDR's weights are reviewed every five years and are based on trade and reserve metrics — a fundamentals-based approach that complements (but does not replicate) the volatility-based approach employed here. A governance framework for the UNIT could usefully incorporate elements of SDR methodology: periodic weight reviews, a clear rebalancing trigger mechanism, and international institutional oversight.

8.3 Limitations

Several limitations of the present analysis warrant acknowledgement. The EGARCH(1,1) specification may be under-parameterised for some currencies (notably BRL and CNY), as suggested by residual autocorrelation patterns in post-fit diagnostics. Higher-order EGARCH specifications or multivariate GARCH models capturing cross-currency correlations could improve both parameter precision and basket weight estimation. The diagonal covariance assumption underlying inverse-variance weighting ignores potentially important cross-currency co-movements that a full mean-variance optimisation would incorporate.

The macroeconomic analysis, while informative, is limited by data availability at annual frequency and the equal-weighting of structural poles in the MVI. A more granular MVI incorporating quarterly or monthly macroeconomic indicators would better align with the daily frequency of the EGARCH analysis. Finally, the 60/40 currency-gold split, while reflecting recent BRICS summit discourse, is not empirically optimised — a formal portfolio optimisation over this allocation parameter could yield a more rigorously justified split.

9. Conclusion

This paper has presented a two-track empirical framework for evaluating the feasibility of a BRICS Currency Unit. Using EGARCH(1,1) models fitted to daily FX data from 2017 to 2024, we estimate time-varying conditional volatility, leverage effects, and shock persistence for each BRICS currency. The resulting inverse-variance weights place the Indian Rupee (46.66%) and Chinese Yuan (39.01%) as the dominant basket anchors, while the Russian Ruble's post-2022 variance explosion effectively excludes it from any financially coherent basket design.

A simulated BRICS Currency Unit index — comprising a 60% EGARCH-weighted currency basket and a 40% gold anchor — achieves an annualised volatility of 13.62% and a total return of +63.39% over the sample period. This performance demonstrates meaningful risk diversification relative to individual BRICS currencies and is broadly competitive with emerging market benchmarks.

The complementary Macroeconomic Vulnerability Index reveals that financial volatility and structural fragility do not always co-move: Russia's pre-2022 macroeconomic fundamentals were stronger than its post-sanctions volatility profile suggests, while Brazil and South Africa exhibit compounding structural vulnerabilities consistent with their higher EGARCH volatilities. India emerges as the single country exhibiting a favourable profile on both short-run financial stability and long-run structural resilience dimensions.

Our findings suggest that a BRICS currency unit is analytically constructible but institutionally challenging. The concentration of basket weight in just two currencies (India and China), the geopolitical tensions among BRICS members, and the demonstrated capacity of sanctions events to destroy the financial stability of individual basket components all pose serious implementation risks. A governance framework incorporating periodic weight reviews, automatic rebalancing triggers, and formal institutional oversight — drawing on SDR methodology — would be essential for any operationalisation of the UNIT concept.

Future research should extend this framework in three directions: (i) multivariate GARCH models capturing cross-currency correlations; (ii) empirical optimisation of the currency-gold weight split; and (iii) incorporating high-frequency macroeconomic data into a dynamic MVI that can be updated in near real-time alongside EGARCH estimates. These extensions would move the BRICS currency unit concept from a research instrument toward an operationally credible policy proposal.

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